Arshia Akhtarkavan

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Education

2023 – Expected 2027 Bachelor of Science, **University of Arizona**, Tucson, AZ (GPA: 4.0/4.0)
W.A. Franke Honors College
Majors: Physics, Mathematics (Comprehensive Emphasis), & Astronomy

2022 – 2023 Diploma, **Saba International High School**, Frankfurt, Germany

2020 – 2022 High School, **The Atomic Energy Institute**, Tehran, Iran

Research Experience

Undergraduate Researcher in Astrophysics Nov 2023 – Present

Advisor: Prof. Eiichi Egami (EGAMI Group) | Steward Observatory, University of Arizona

Galaxy Kinematics and Morphology

Research: Investigate the kinematics and morphology distant emission-line galaxies using JWST/NIRCam slitless spectroscopy. Developed and applied forward models using orthogonal grisms to infer galaxy rotation curves and morphology.

Outcomes: First-author manuscript in preparation; presented findings at AAS 247 and other conferences.

High-Redshift Transients and Variability

Research: Searched multi-epoch JWST/NIRCam imaging for transient and variable sources, identifying and characterizing candidates in distant galaxies.

Outcomes: Coauthored a peer-reviewed publication on a $z > 5$ transient/variable source (*ApJ*, 2025) and a TNS AstroNote reporting 39 transients/variables (2024).

Additional contributions: Processed JWST/NIRCam imaging and slitless spectroscopy; developed deep-learning methods for resolving ambiguous spectroscopic redshift solutions; contributed to the SAPPHIRES Early Data Release (*ApJS*, accepted).

Dorrit Hoffleit Research Scholar Jun – Aug 2026

Advisors: Prof. Marla Geha (Geha Group) & Prof. Earl Bellinger (YAML Group) | Yale University

RR Lyrae Stars in the Draco Dwarf Galaxy

Research: Investigated the properties and dynamics of RR Lyrae stars in Draco using spectroscopy and stellar-evolution modeling. Developed Bayesian methods for radial-velocity inference and distance estimation, aiming to constrain the galaxy's internal dynamics.

Outcomes: Presented research at two Yale workshops and symposia; analysis ongoing toward a research manuscript.

Researcher in Mathematical Physics Oct 2024 – Jan 2026

Mathematics Department, University of Arizona (Advisor: Prof. Sergey Cherkis)

The Islands Project — Nahm Equations

Research: Investigated the mathematical structure of Nahm equations, focusing on the construction of higher-order solutions and connections to gauge theory. Also rebuilt the Islands Project website to support student engagement in mathematical physics research.

Outcomes: Produced an independent research report, presented the project at a university research forum, and received a competitive undergraduate research grant.

Publications

1. Forward-Modeling Galaxy Kinematics and Morphology with Orthogonal JWST/NIRCam Grisms in the SAPPHIRES Early Data Release
Akhtarkavan A., Liu Y., Egami E., Sun F., Zhu Y., et al. (manuscript in preparation).

2. The First Photometric Evidence of a Transient/Variable Source at $z > 5$ with JWST
DeCoursey C., Egami E., Sun F., **Akhtarkavan A.**, et al. (2025, *ApJ*, 990, 31; arXiv:2504.17007).

3. Slitless Areal Pure-Parallel HIgh-Redshift Emission Survey (SAPPHIRES): Early Data Release of Deep JWST/NIRCam Images and Spectra in MACS J0416 Parallel Field
Sun F., Fudamoto Y., Lin X., Helton J. M., Hsiao T. Y-Y., Egami E., **Akhtarkavan A.**, et al. (*ApJS*, accepted; arXiv:2503.15587).

4. Discovery of 39 Transients/Variables ($z_{\rm host} = 0.437$–5.271) in GOODS-N using JADES and CONGRESS JWST/NIRCam Images
DeCoursey C., Egami E., Sun F., **Akhtarkavan A.**, et al. (2024; Transient Name Server AstroNote 264).

Conferences and Workshops

1. **"Forward-Modeling Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
Flash Talk and Poster Presentation, *Surveying the Universe in 4D: Beating Cosmic Variance with Wide-Field Slitless Spectroscopy from HST, JWST, Euclid, Roman, and Beyond* Workshop
Space Telescope Science Institute (STScI), Baltimore, MD, August 2026

2. **"Modeling RR Lyrae Spectra in the Draco Dwarf Galaxy"**
Contributed Talk, *Yale/CeNAM Nuclear Asteroseismology Workshop*
Yale University, New Haven, CT, July 2026

3. **"Modeling RR Lyrae Spectra in the Draco Dwarf Galaxy"**
Contributed Talk, *Summer Research Symposium*
Yale University, New Haven, CT, July 2026

4. **"Islands Project and the Nahm Equations"**
Talk, *Third Annual Cross-Campus Undergraduate Research Forum and Questionarium*
University of Arizona, Tucson, AZ, April 2026

5. **"Forward-Modeling Emission-Line Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
Contributed Talk, *Steward Observatory Internal Symposium*
Steward Observatory, University of Arizona, Tucson, AZ, March 2026

6. *Arizona Winter School on Multi-Probe Cosmology in the Roman ST Era*
Arizona Cosmology Lab, University of Arizona, Tucson, AZ, January 2026

7. **"Forward-Modeling Emission Line Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
iPoster presentation, *247th Meeting of the American Astronomical Society (AAS 247)*
Phoenix, AZ, January 2026

Awards and Scholarships

Dorrit Hoffleit Research Scholarship, Yale University (2026)
Evelyn O. Bychinsky Promising Astronomer Award (2026)
William F. Lucas Astronomy Scholarship, Steward Observatory (2026)
College of Science Galileo Circle Scholar (Mathematics), University of Arizona (2026)
College of Science Galileo Circle Scholar (Physics), University of Arizona (2026)
Weaver Award for Undergraduate Research (2026)
Kenneth S. Krane Physics Scholarship (2025)
Evelyn O. Bychinsky Promising Astronomer Award (2025)
RII-Sponsored Campuswide Undergraduate Student-Initiated Original Research (2025)

Glenn C. Purviance Scholarship in Physics (2023)
Academic Year Highest Academic Distinction (2023 – Present)
Dean's List with Distinction - Full Time (2023 – Present)
Global Wildcat Tuition Award, University of Arizona (2023 – Present)
Silver Medal, National Astronomy and Astrophysics Olympiad (2022)

Independent Studies

Quantum Field Theory Fall 2026, in progress

Physics Department, University of Arizona (Advisor: Prof. Meinel)

– Graduate-level independent study

– Completing weekly problem sets and a final-term project

Nahm Equations and Mathematical Physics Fall 2025

Mathematics Department, University of Arizona (Advisor: Prof. Cherkis)

– Studied Nahm equations and related topics in gauge theory and representation theory

– Wrote a final report on the construction of higher-order Nahm solutions

Professional Experience

Astronomical Software Engineering Intern Aug 2026 – Present

TIMESTEP, University of Arizona (Event Horizon Telescope Group; Supervisor: Dr. CK Chan)

– Participating in an academic-year internship focused on developing `Hallmark`, a lightweight Python package for version-controlling and distributing scientific data products across Event Horizon Telescope analysis pipelines.

Mechanical Engineering Workshop Intern Jan – Mar 2023

FH Aachen University, Essen, Germany

– Gained hands-on experience operating machining and workshop equipment and interpreting and producing technical schematics.

Teaching and Outreach

Mathematics Grader Aug 2025 – May 2026

Mathematics Department, University of Arizona, Tucson, AZ

– Graded assignments and examinations, prepared detailed solutions, and provided written feedback for upper-division courses in differential equations and linear algebra.

Undergraduate Mentor Oct 2026 – Present

University of Arizona, Tucson, AZ

– Participate in an undergraduate mentoring program, providing peer guidance and support to one student.

Astronomy Olympiad Teaching Assistant Oct 2022 – Aug 2023

The Atomic Energy Institute, Tehran, Iran

– Taught cosmology and prepared lecture notes for Astronomy Olympiad students, based on textbooks by Ryden and Carroll.

– Selected problems and led problem-solving sessions in celestial mechanics.

Educational Materials Translator 2025

Arizona Astrobiology Center, University of Arizona, Tucson, AZ

– Translated Astrobiology lecture materials from English into Farsi for Persian-speaking elementary school students

Skills

Programming Languages	Python, C/C++, SQL
Scientific & Data Analysis Libraries (Python)	NumPy, Matplotlib, PyTorch, Astropy, SciPy, Specutils, Photutils, Pandas, JAX
Astronomy & Astrophysics Software	DS9, JWST Pipeline, Source detection and Photometry, Webbpsf, NIRCam imaging and grism spectroscopy
Software & Research Systems	Linux/Unix, Git/GitHub, HPC environments, remote computing (ssh), GPU-accelerated computing (CUDA), Jupyter, VS Code, Emacs, LaTeX, Microsoft Office
Concepts	Data Processing Pipelines, Scientific Computing, Machine Learning, Numerical Methods

Languages

English	Fluent
Farsi (Persian)	Native
German	B1 level
Mandarin Chinese	Currently learning